

The ETF Premium Strategy

Running a good volatility trading book/business means that you have a clean way to extract the variance risk premium. And the best way to do this is by selling options on ETFs.

The strategy is simple. We sell options that meet the following criteria:

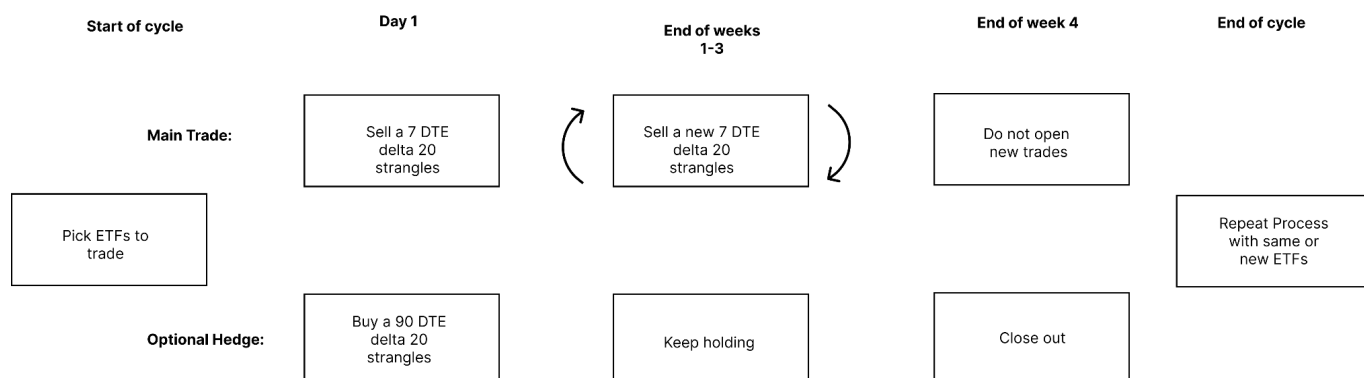
1. The underlying is an ETF.
2. Has a positive average variance risk premium.
3. Has a positive average short straddle return.
4. Has an IV percentile lower than 95.
5. Each ETF in the portfolio represents a different industry or market.

The execution is also simple:

1. We sell weekly delta 20 strangles for 7 days and roll them for 4 weeks before rotating to a new ticker. We do not delta hedge unless there is a massive jump.
2. If we want to hedge our downside risk, we can roll the weekly strangles under the protection of a 90 DTE delta 20 strangle.

This basket makes money. It outperforms. And it's really low maintenance.

Here's the entire strategy laid out for you.



The scanner and ETF page make step 1 easy (it shows us the best ETFs for this strategy and the backtests of their performance). All you need to do from there is pick the ETFs and place the trades.

That's it. I wasn't lying when I said this was going to be boring.

But for the sake of being diligent individuals, let's take 10 minutes to look at the performance and understand all the details about how this gets implemented.

I mean after all, we aren't going to have much to do after we understand it, so we might as well put in *a little bit of effort* now, right?

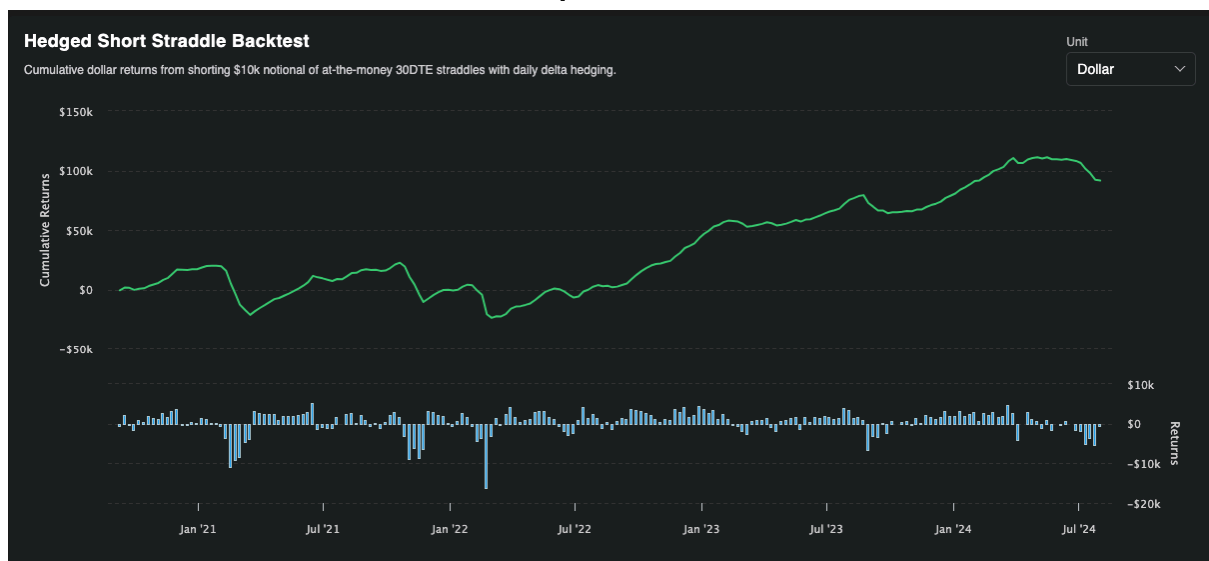
ETF Strategy performance

In this section we are going to go over the strategy performance. We are going to talk about the returns, drawdowns, and variance. Basically, after this section you will have a benchmark for performance so you can know if you are doing things correctly and there should be no surprises for you in the future (during upswings or downswings!)

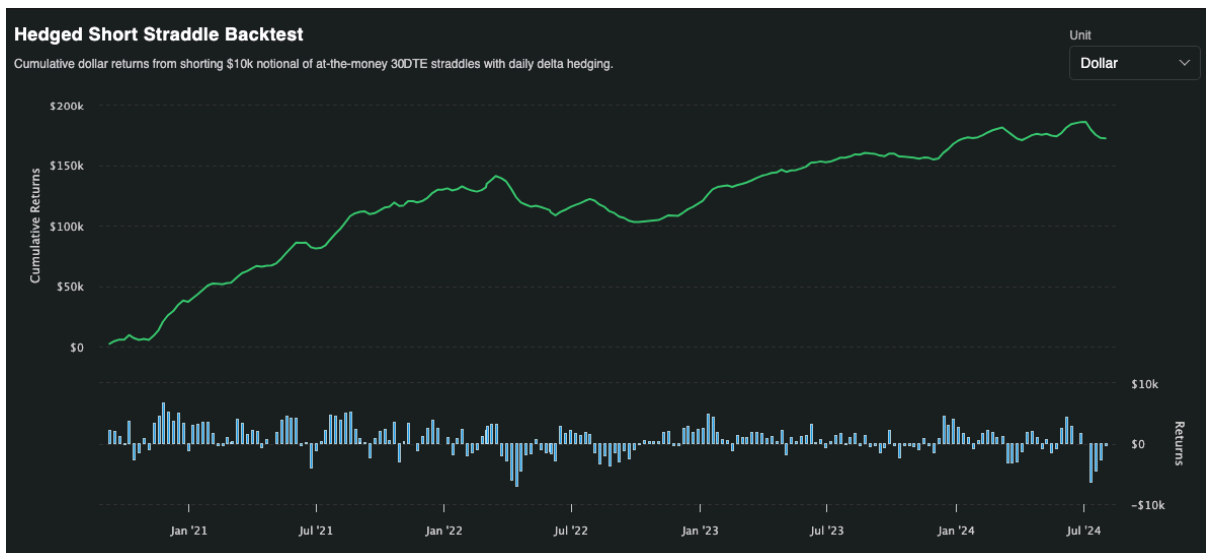
To start, here is a backtest of the performance of selling options on some of the major ETFs to show how across industries and sectors, the variance risk premium exists.



\$SPY performance



\$USO performance



\$DIA performance

As you can see, the performance is solid across the board. From ETF to ETF, there are differences in the shape of their returns, PnL variance, drawdowns, etc. Looking at the PnLs above you can see how (1) they all have a positive return and (2) they all take a different path to achieving these returns.

When one makes money, maybe another one is more flat. If we were to have all of these trades in a basket together (or even better ones) then we can see how our *average PnL in our portfolio* should be very smooth and balanced.

Not bad for an hour of work per week. Individually selling options on the ETFs makes money, and by diversifying across a basket of the ones with a positive risk premium we can smooth our PnL and achieve a highly efficient return.

Note: you will have access to a trade scanner based on the strategy criteria and real time backtests like the ones above for every ETF, so don't worry, you don't need to guess which ETFs to trade.

Now that you've seen the performance, let's dive into how this strategy works.

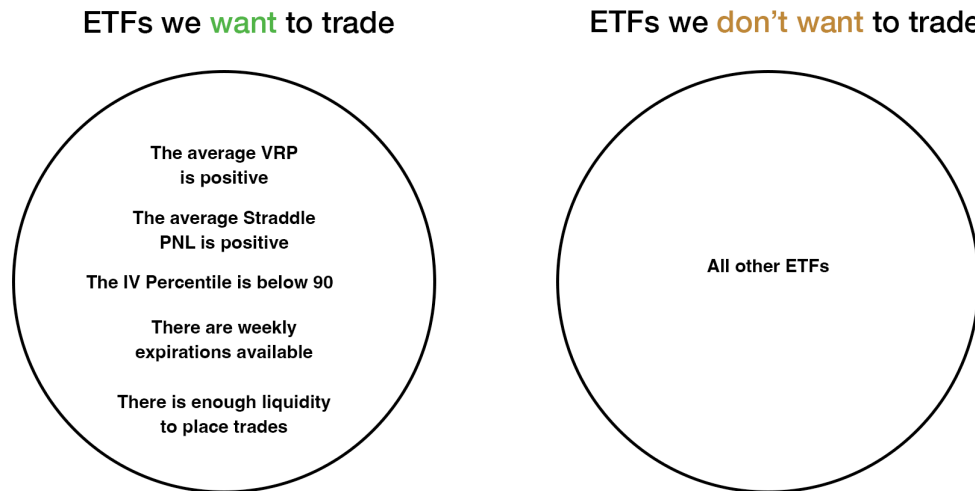
To successfully run this strategy, we have one core question to answer:

Which ETFs have a positive variance risk premium?

You can think of this as sorting ETFs into two buckets. The "*Haves*" and the "*Have Nots*".

Those that "*Have*" a variance risk premium are fair game for our portfolios. The ones that do not are removed from our analysis.

We use the VRP analysis to sort ETFs into two groups:



**Every ETF falls into one of these two buckets.
All the ones we want to trade are good candidates.**

This strategy uses 4 factors to find profitable trades

1. We only trade ETFs.
2. They must have a positive variance risk premium.
3. They must have a positive average straddle return.
4. Their IV Percentile must be below 95.

Let's go over each of them briefly:

(1) Why do we only trade ETFs?

There are 3 major reasons :

1. ETFs have added premiums baked in because of correlation risk (all the holdings become correlated in a crash). In fact it has been argued that the majority of the variance risk premium is due to this correlation risk.
2. ETFs are a common place for funds to hedge their book (good liquidity and sector exposures)
3. ETFs don't carry the same bankruptcy and upside jump risk that individual stocks do (less blow up risk)

(2) Why must there be a positive VRP?

I am hoping this one is self explanatory by this point! If it's not, go back to read the chapter on variance risk premium. For good measure, here's the clarification: We utilize the mean reversion characteristic of VRP to indicate if an ETF has the risk premium. Basically, if the average VRP is positive, then on average, we should capture a positive return from selling options on this ETF. The way we measure the VRP is by looking at the average return selling 30 day straddles over the last 4 years.

(3) Why must the average straddle return be positive?

The reason the average straddle return must be positive is because it is the metric that tells us if we are actually able to monetize the variance risk premium. This metric takes into account trading costs, slippage and delta hedging. All of the things that impact the *practicality* of capturing the risk premium through the options we sell. By seeing an average straddle return, we know that after costs and delta hedging, we are able to extract a positive return on average.

(4) Why must the IV Percentile be below 95?

The reason we only trade ETFs that have an IV percentile below 95 is because we want to control how much our PnL swings and reduce the risk of large drawdowns. Trading when the level of implied volatility this high results in greater PnL variance without additional compensation. The table below illustrates this point:

VIX Percentile	Lower VIX Level	Average VP (points)	Average VP (as %)
0.9<Max	27.0	6.7	24.8
0.8<0.9	22.4	6.7	30
0.7<0.8	19.5	5.2	26.5
0.6<0.7	17.3	4.2	24.4
0.5<0.6	15.9	3.8	24.2
0.4<0.5	14.4	2.5	17.6
0.3<0.4	13.1	1.3	10.0
0.2<0.3	12.3	3.1	25.3
0.1<0.2	11.1	2.3	20.8
0<0.1	9.1	2.5	27.4

This table shows the variance risk premium at different VIX percentiles (remember, VIX is the implied volatility for the S&P, or what we would use as a reference point for “market volatility”).

In the table you can see that there is a relationship where as the percentile increases, the average risk premium in points also increases. At first you might think “*Wait a second! Higher percentile means higher risk premium!*”. But when we normalize the data, we see that *the risk premium as a percentage of the implied volatility* is actually pretty constant (the last column of the table).

We know that at higher levels of volatility, our strategies experience higher volatility. We also know that the risk premium is present regardless of the level of volatility.

So by *not trading* when the IV percentile is high (above 95), we reduce our PnL variance without sacrificing the risk premium since it will still be there when things calm down!

3 Steps to add a new trade to your portfolio

Most traders will continue to add ETFs to their portfolio as they become more confident in the strategy and opportunities come up. These 3 steps are what we do every time we want to add a new ETF to our portfolio:

1

Scan for ETFs that are “haves”

- Done for you: ETF Premium Scan
- All ETFs meet the “good trade” criteria
- Start with 2-3 ETFs to trade
- Pick ETFs that represent different sectors/regions

2

Pick the ones we will add to our book

- Use correlation matrix to see relationship between the ETFs you selected
- If high correlation, pick new ETFs.
- If stuck choosing between two ETFs, pick one with higher liquidity

3

Sell 7 DTE Delta 20 Strangles

- Optional hedge: buy a 90 DTE delta 20 strangle
- Straddles are OK too, since they are delta neutral
- Selling DTE less than 30 is OK too

See how this only takes *maybe* an hour a week? It's boring, but it's pretty cool to think the strategy that gets us 100% of the way to our goal is actually not a full time job to run.

So yea, as promised, **it's a boring strategy that actually makes money.**

Up next: how to find and pick your trades (in detail)